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Unsupervised summarization of arguments toward key point generation with Sentence-BERT-based method

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Abstract Many datasets with large number of arguments are used in the field of argument mining. To make a concise summary (key point) from a set of arguments in such datasets, two basic tasks are necessary: match scoring, a step to cluster arguments into similar groups; and key point generation. The previous studies focus on clustering arguments to key points for the first step. However, those studies utilize manually created key points which is time-consuming and costly. To address this problem, we experiment with the classical summarization baselines. We also propose an unsupervised approach for summarizing arguments. In our previous work, we show the Sentence-BERT-large model is most accurate for arguments without labelled data. Therefore, we summarize arguments in our created data with our proposed method. We apply a key phrase extraction method, i.e. EmbedRank, to key point generation using Sentence-BERT. Experimental results show the usefulness of Sentence-BERT for summarizing arguments.

Keywords Natural Language Processing, Argument Mining, Sentence-BERT, Summarization, Debates

1 Introduction

Recently, the area of argument mining has become popular in the field of NLP [1, 2]. This topic spreads from extracting argument components (e.g. claims, premises), predicting persuasiveness of an argument [3, 4], to filling gaps between claims [5, 6], and predicting argument stance (pro/con) [7]. While there are numerous studies on arguments from various perspectives, no method will be practical if users cannot find accurate opinions for their debating topic. Then, how to gather such accurate arguments? There are typically two approaches to collecting arguments; one way is to retrieve them from online debate resources (like *iDebate* or *CreateDebate*), another is to gather arguments from debate spectators who express their opinions [2]. However, both methods have a common problem: it is difficult to read and understand all arguments about a single debate topic because their numbers often exceed hundreds. Reading through such amount of text would be very laborious. In order to solve this problem, Reimers et al. [8] use capabilities of contextualized word embeddings of ELMo [9] and BERT [10] to classify and cluster topic-dependent arguments from Argument Facet Similarity Corpus [11]. They clustered related arguments, however they did not generate summaries of arguments. This problem is addressed in Bar-Haim et al. [12]. In their paper, they defined the key point: usually single sentence describing a set of similar arguments. They generate a large dataset consisting of [argument, key point] pairs. Next, they conduct two experimental steps: *Match Scoring* and *Match Classification*. In *Match Scoring*, they compute a match score for a given [argument, key point] pair, (see Table 1, which shows examples of the data for *Match Scoring*, named ArgKP). In *Match Classification*, their methods discover matching key points for each argument. Although they claim that their research utilizes argument summarization, only the relevance between key points and arguments is calculated, which means their both experiments are almost identical to the above-mentioned research [8]. Furthermore, for *Match Scoring* and *Match Classification*, creating key points is necessary beforehand. This indicates that their methods cannot be used with new debate topics for which key points do not exist. For mapping arguments to key points, they also point out that BERT-large [10] fine-tuned model achieves the best results with supervised learning, and a BERT-large embedding method yields the best results among unsupervised methods. Our previous research [13] showed that Sentence-BERT-large model [14] provides more useful embedding model than other methods in order to embed arguments and key points for *Match Scoring* described above. In that previous work, we assume that one of important factors for summarizing large number of arguments into one key point is to map an argument to the key points, not to

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Topic: **Homeschooling should be banned**

Arguments	Key Points	Stance	Match
education at home could represent a risk since it is not regulated	Homeschools cannot be regulated/standardized	1	1
a parent knows best what is good for their kid	Parents should be permitted to choose the education of their children	-1	0

Table 1: Example of [argument, key point] pairs on a given topic. Stance and Match refer to pro (1) or con (-1), and same meaning (1) or not (0), respectively.

select a key point from one argument. Therefore, we adopt the *Match Scoring*, not the *Match Classification*, utilizing Sentence-BERT.

In this paper, we propose a new task, *Argument Summarization*, for generating a key point from a set of arguments. We summarize arguments utilizing the ArgKP dataset [12], which is explained in Section 3.2, to generate a key point. To the best of authors' knowledge, this is the first attempt of this kind. We utilize the classical, but widely used standard unsupervised models for baselines. Because the size of the dataset is too small for supervised methods, and our previous research shows Sentence-BERT-large model is useful for the ArgKP dataset, we decided to utilize a Sentence-BERT-based unsupervised approach.

The main contributions of this article are:

1. Exploring a new task for *Argument Summarization* toward generating key points for arguments, and providing baselines using various classical unsupervised summarization methods;
2. Proving that Sentence-BERT is useful for summarizing argument when compared with another embedding model;
3. Introducing a new method to extract the exact argument for a key point, i.e. EmbedRank-based method which is based on Sentence-BERT model.

2 Related Works

Argument mining has started with researchers working on argument components for debate topics, discussions or student essays. There are several works on argument persuasiveness [3] or persuasiveness of the whole debate [15] for generating more convincing arguments. In order to conduct research on arguments from several perspectives, various large argument datasets have been recently created [16, 17]. Stab et al. [16] also focus on retrieving arguments from different sources. Their dataset covers about 90% of arguments found in expert-curated lists of arguments from an online debate portal. Similarly, the above-mentioned ArgKP dataset conducted by Bar-Haim et al. [12] also indicates high agreement between expert dataset (key point list) and arguments, achieving Cohen's kappa=0.82. More recently, several researchers have been working on argument facets to identify whether two arguments are similar or not [11, 18]. Others concentrate on examining whether these two arguments have the same facet or not [19]. Their research is similar to mapping arguments to a key point, but they do not calculate similarity between a facet and an argument. Moreover, they do not generate or clarify key points, and in this aspect, their work clearly differs from ours. As for research on mapping arguments to key point, existing research [20, 12] can be given as the most known examples. Boltužić and Šnajder [20] implement argument clustering. They map arguments derived from one online debate portal (*ProCon*) to arguments of other portal (*iDebate*). However, they create only two debate topics with less than 400 arguments. Bar-Haim et al. [12] make a massive dataset including pairs of an argument and a key point labelled with relevancy. They also propose *Match Scoring* and *Match Classification* for the first steps to summarize arguments. *Match Scoring* would be valid for summarization, but *Match Classification* might not be useful for summarizing arguments due to the reasons described in Section 1. Therefore, they do not summarize arguments nor extract any important argument from the argument list. Our work differs from theirs in this aspect.

3 Experimental Setup

In this section, we describe our preparations for *Argument Summarization* task: the task setting, the data for the task, and its evaluation metrics.

3.1 Argument Summarization

In *Argument Summarization*, we summarize a set of similar arguments and generate a key point. We create data

for *Argument Summarization* basically from ArgKP dataset [12], and try to extract the most accurate argument from a set of similar arguments in order to generate a key point. Finally, we evaluate the output with two evaluation metrics.

3.2 Data

We utilize ArgKP dataset for *Argument Summarization*. In order to adapt ArgKP to this task, first we delete the pairs without agreeing arguments and key points, and group arguments into the same key points. As an example for [key point, arguments] pair groups, Table 2 shows a set of arguments which are given for the key point "Parents should be permitted to choose the education of their children" in the debate topic "Homeschooling should be banned". ArgKP also includes some arguments which were paired with more than one key points because they indicate two or more facets. In the dataset, a [key point, arguments] pair group is counted as one data sample. We divide the dataset into 21 train topics (182 groups), and 7 test topics (61 groups).

3.3 Evaluation Metrics

For evaluation, we used two summary evaluation metrics; ROUGE [21] score, and BERTScore [22]. ROUGE is a recall-based metric for fixed-length texts which is based on n-gram co-occurrence. BERTScore computes similarity scores between each token in the original and generated summaries. This metric is more robust to difficult problems such as paraphrases that occur between arguments implying the similar content. Therefore, we use BERTScore in addition to ROUGE. We evaluate our methods with F1 for ROUGE-1, ROUGE-L and BERTScore.

4 Baselines

For *Argument Summarization*, we adopt LexRank and TextRank as classic, but strong extractive baseline methods as comparison with our proposed method.

4.1 LexRank

LexRank was initially proposed by Erkan and Radev [23]. It is a graph-based extractive summarization method for computing relative importance of texts. It constructs a sentence graph whose edge weights (0 / 1) are decided from a cosine similarity. Using LexRank, we extract the highest ranked argument from all arguments. The threshold to decide the edge weights is 0.3, which maximize F1 of BERTScore on the train data.

4.2 TextRank

Page et al. [24] proposed TextRank, a ranking algorithm based on PageRank [25], which is often used in keyword extraction and text summarization. In order to find relevant keywords, TextRank constructs a word/sentence network by looking which words follow one another. A link is created between two words if they follow one another, and link weight increases if these two words occur more frequently adjacent to each other. With TextRank, we extract the highest ranked argument.

5 Proposed Method

This section describes how to utilize EmbedRank to Argument Summarization.

5.1 EmbedRank

This section describes EmbedRank, which we used for summarizing arguments in the previous work. EmbedRank is an unsupervised method with document embeddings for key word extraction [26]. We adapt EmbedRank, which can be used for key words extraction, to order to extract sentences. Moreover, to obtain more accurate sentence embeddings, we propose to use Sentence-BERT [14] instead of Doc2vec [27] and Sent2Vec [28] which were used by [26].

In EmbedRank, there are two parameters; *key size* and λ. *Key size* defines the number of extracted key phrases. When *key size* equals 1, only one key phrase is extracted by EmbedRank. λ is a parameter in the equation for Maximal Marginal Relevance (MMR) [29] which is introduced in EmbedRank. MMR controls the diversity of extracted words and the relevance between documents and extracted words. With a given input query Q, the set S represents documents that are selected as correct answers for Q. S is populated by computing MMR as described in Equation (1),

$$MMR := \arg\max_{D_i \in R \setminus S} \{\lambda * Sim_1(D_i, Q) - (1-\lambda) \max_{D_j \in S} Sim_2(D_i, D_j)\} \quad (1)$$

where R is the ranked list of documents retrieved by an algorithm, S represents the subset of documents in R which are already selected, D_i and D_j are retrieved documents, and Sim_1 and Sim_2 are similarity functions. The smaller λ becomes, the diversity of outcomes increases.

5.2 Sentence-BERT

Siamese Neural Networks [30] consist of two neural networks trained with the same weights. These weights are trained on two different input vectors. Reimers and Gurevych proposed Siamese BERT Network and implement-

Topic: **Homeschooling should be banned**

Key Point: **Parents should be permitted to choose the education of their children**

ID	Arguments
1	a parent knows best what is good for their kid
2	home schooling is a good alternative for many. it is economical, safe and and a better environment for some children. parents have a right to choose what is best for their child.
⋮	
27	the parents should be allowed to home school their kids if that's their choice.
28	this is a free country, people have the right to homeschool their children

Table 2: Example of a key point describing a set of arguments on a given topic

ed it as Sentence-BERT[1] (SBERT for short) [14] trained on Natural Language Inference (NLI), SNLI and MultiNLI datasets, in order to create universal sentence embeddings. Following the original Sentence-BERT [14], we use mean-pooling model.

There are other kinds of Sentence-BERT models, i.e. Sentence-BERT-STSb. SBERT-STSb model is also mentioned in the same work [14]. These models are first fine-tuned on the AllNLI dataset, then on the train set of STS benchmark. With this fine-tuning phrase of training process, SBERT-STSb is well suited for measuring semantic textual similarity.

In our proposed model for *Argument Summarization*, we adopt Sentence-BERT-large model because our previous work shows this model is the best performing one.

5.3 Our Proposed Method

First, we set the parameters and input texts as below. The smaller λ becomes, the diversity of outcomes increases, therefore we set λ to 1. Using cosine similarity for the similarity functions, we compute MMR score between an argument *Arg* and a key point *KP*. In our method, MMR score is calculated with the following Equation (2):

$$MMR = \underset{Arg_i \in A \setminus K}{\arg\max}\{cos_{sim}(Arg_i, KP)\}, \quad (2)$$

where A is the set of candidate arguments, and K is the set of extracted key points. A key point KP is the label for our dataset, therefore we average the embeddings of the input arguments, and adopt the averaged embeddings instead of KP. In order to obtain MMR score, Bennani-Smires et al. [26] computed embeddings of documents with Doc2vec and Sent2Vec. In our *Argument Summarization*, we use Sentence-BERT-large, which is the best performing *Match Scoring* embedding method, in addition to these methods. This is because Bennani-Smires et al. [26] used EmbedRank for documents, e.g. the Inspec dataset, and NUS, whereas our target is [key point, arguments] pair groups, which in most cases include only one sentence. As the Sentence-BERT achieves highest scores in many downstream tasks [14], we decided to adopt it. For comparison, we also compute the summarization evaluation scores with Doc2vec-based EmbedRank[2]. The parameter *key size* is set up to 1 in order to obtain the highest ranked argument.

	R1	RL	BERT
LexRank	0.126	0.126	0.872
TextRank	0.126	0.118	**0.880**
EmbedRank w/ Doc2Vec	0.102	0.101	0.868
EmbedRank w/ SBERT-large	**0.132**	**0.131**	0.879

Table 3: F1 score of ROUGE-1, ROUGE-L and BERTScore results of argument summarization. R1, RL, BERT refer to ROUGE-1, ROUGE-L, and BERTScore, respectively.

6 Evaluation Results

Table 3 shows comparison of ROUGE scores and BERT-Score for various *Argument Summarization* methods evaluated on the test data. As shown in Table 3, our method, Sentence-BERT-based EmbedRank, exceeds classical methods considering ROUGE scores, however our approach achieves almost the same BERTScore as TextRank. This means we cannot claim that our method yields a significant differ-

[1] We use Sentence-BERT model available at `https://github.com/UKPLab/sentence-transformers`

[2] We use the pre-trained Doc2vec model trained on the English Wikipedia corpus. This model is available at `https://github.com/jhlau/doc2vec`.

ence from other methods. Regarding the usage of Sentence-BERT, we can say it is efficient for *Argument Summarization* in addition to *Argument Scoring* because the results of Sentence-BERT-based EmbedRank are superior to Doc2vec-based one except for precision of ROUGE-1.

The reason of the small values for ROUGE-1 and ROUGE-L is that there are some cases that several output arguments have the same meaning to the key point (reference) but only few overlapping words between the key point and those arguments.

7 Error Analysis

As can be observed from Table 4, BERTScore in our method shown in Table 4 is lower than others. However, an argument extracted by Doc2vec-based method (see Table 2) seems to contain another semantic opinion, i.e. it is labelled as similar to two key points: "Homeschools can be personalized to the child's pace/needs" and "Homeschooling is often the best option for catering for the needs of exceptional/religious/ill/disabled students." Even BERTScore cannot evaluate such complex texts, so we should consider introducing the human evaluation. This problem may have negative impact not only on evaluation, but also on extraction. To avoid these cases, we consider working on argument facets in the near future.

8 Conclusion

Our work is the first attempt to summarize arguments to one key point. We proposed a new task *Argument Summarization*, whose purpose is to generate a key point from a set of similar arguments.

For the first step of such key point generation, we focused on the existing task, *Match Scoring*, and our previous work [13] which showed that Sentence-BERT-large is the best embedding model for the ArgKP dataset [12]. Then, we utilized Sentence-BERT-large model to *Argument Summarization*. We utilized the ArgKP dataset for *Argument Summarization* task, and extracted a key argument using classical but widely used methods, i.e. LexRank and TextRank. In addition to the standard methods, we also applied Sentence-BERT-based EmbedRank to the *Argument Summarization* task. We also used Doc2vec-based EmbedRank for comparison. As a result, Sentence-BERT-based EmbedRank did not show significant improvement, however Sentence-BERT-based method outperformed Doc2vec-based one. For the future work, we will resolve a problem of arguments with two or more opinions, as they negatively influenced our results.

Finally, we plan to add the scores, which measure argument persuasiveness, to the MMR score described by Equation (2). Whether or not the extracted arguments are sufficiently persuasive is directly related to the impression we get from them. If the argument is not persuasive enough, debate audience will not be convinced. However, if the opposite is true, they will be convinced. For this reason, we are working on scores for measuring an argument persuasiveness according to our previous research on debate outcome prediction [15].

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Topic: **Homeschooling should be banned**

Methods	Generated Key Points	*BERT*
Original	Homeschools can be personalized to the child's pace/needs	None
LexRank	homeschooling can provide more one on one attention and can allow the child to learn at their own speed.	0.902
TextRank	homeschooling allows parents to teach using methods that are best suited to their child's way of learning	0.903
EmbedRank w/ Doc2Vec	homeschooling provides an opportunity to tailer the curriculum to the particular childs learning style. this may be particularly advantageous when individual learning needs require special consideration.	0.902
EmbedRank w/ SBERT-large	some children are better suited to the homeschool environment and learn better that way.	0.874

Table 4: Generated key point example compared to an original key point. *BERT* refers to F1 in BERTScore between an original key point and a generated key point.

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